Subject Description Form

Subject Code	LSGI537
Subject Title	GIS for Humanities and Social Sciences
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. Introduce the interdisciplinary nature of Geographic Information Science (GIS) and its integration of GIS tools and methods into research fields of Humanities and Social Sciences. 2. Enlighten students to analyse historical events and convert the textual information into digital format by employing GIS, geoprocessing tools, and statistical techniques for spatiotemporal analysis. 3. Enable students to combine knowledge on the human-nature linkages in the past with geographical scale thinking in time and space based on the empirical results by GIS and related quantitative tools. 4. Equip students with the knowledge and ability to scientifically re-interpret distribution of cultural/historical heritages, contemporary human communities, and related sustainable strategies from the perspective of human-nature linkages in the past.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1. Critically assess and incorporate qualitative data into GIS analyses to enrich interpretations of historical and social phenomena by text mining historical documents. 2. Utilise GIS and quantitative techniques to scientifically analyse spatial and temporal patterns of natural phenomena, and social events, which are narratively recorded in historical documents. 3. Obtain the knowledge on both physical and human geography in a historical and cultural context, such as climate change, natural disaster, human migration and war, by empirically analysing historical cases in a large amount. 4. Develop the geographical scale thinking in time and space by focusing on different lengths of historical periods (for example, decadal, centennial, millennium) and diverse spatial coverages (for instance, city, nation and continent). 5. Enable a view on human-nature linkage from the past to explain the current distribution of cultural/historical heritages, to re-examine the vulnerabilities of modern human communities, and to review related sustainable strategies under various environmental challenges.
Subject Synopsis/ Indicative Syllabus	The subject introduces students to the interdisciplinary integration of GIS and research of Humanities and Social Sciences. It explores how GIS tools enable the digitisation, analysis, visualisation, and management of textual information from historical documents. In addition to methodological training, the theoretical discussion based on spatiotemporal analysis by GIS will equip the students with knowledge and scale thinking by studying physical and human geography in the past to empirically review the modern sustainable strategies of urban development in the world and explain the current distribution of cultural/historical heritages.

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<tr><td>Teaching/Learning Methodology</td><td>
1. Lectures and invited talks (ILOs 1, 3, 4, and 5): provide foundational knowledge on the features of historical documents as well as the past major patterns of human-nature linkages at various scales in time and space by analysing historical cases with different themes, such as climate change, natural disaster, human migration and war.
2. Tutorials and lab work (ILOs 1, 2, 3 and 4): focus on using GIS software to perform key tasks such as digitisation, georeferencing, geocoding, and database management by using records of historical documents from different dynasties and regions.
3. Individual project for presentation and written report (ILOs 1, 2, 3, 4, and 5): analyse a real issue by collecting and using historical records and present findings using GIS visualisation and quantitative tools.
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<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes</td><td>
<table>
<tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="5">Intended subject learning outcomes to be assessed</td></tr>
<tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr>
<tr><td>1. Participation</td><td>20%</td><td>√</td><td></td><td>√</td><td>√</td><td>√</td></tr>
<tr><td>2. Individual project presentation</td><td>30%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr>
<tr><td>3. Individual project report</td><td>50%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr>
<tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr>
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Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:
Participation: Students are required to attend the lectures. Furthermore, they should be involved in class discussions and share their views on the presentations by other classmates in this subject.
Project Presentation and Project report: Students analyse a real-world example of GIS application, focusing on spatial and temporal data relationships, visualisation techniques, and integration of qualitative data. Results are submitted in both oral presentation with a visualisation format and written report with a quantitative support.
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<tr><td>Student Study Effort Expected</td><td>
<table>
<tr><td>Class contact:</td><td></td></tr>
<tr><td>▪ Lecture (including invited talk)</td><td>26 Hrs.</td></tr>
<tr><td>▪ Tutorial (including lab works)</td><td>13 Hrs.</td></tr>
<tr><td>Other student study effort:</td><td></td></tr>
<tr><td>▪ Self-study, reading and practical</td><td>78 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>117 Hrs.</td></tr>
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<tr><td>Reading List and References</td><td>
Gregory, I. N. (2003). A place in history: A guide to using GIS in historical research. Oxbow Books.
Gregory, I., & Ell, P. S. (2007). Historical GIS: Technologies, methodologies, and scholarship. Cambridge University Press.
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	Paul A. Longley, Michael F. Goodchild, David J. Maguire, David W. Rhind (2015) Geographic Information Science and Systems 4th Edition; Wiley Kang-tsung Chang (2018) Introduction to Geographic Information Systems; McGraw Hill Michael Law & Amy Collins (2022) Getting to Know ArcGIS Desktop 10.8; Esri Press Burrough, P. A., McDonnell, R. A., & Lloyd, C. D. (2015). *Principles of geographical information systems*. Oxford University Press, USA. Conolly, J., & Lake, M. (2006). *Geographical information systems in archaeology*. Cambridge University Press. Gregory, I. N., & Geddes, A. (Eds.). (2014). *Toward spatial humanities: Historical GIS and spatial history*. Indiana University Press. Knowles, A. K. (2005). Emerging trends in historical GIS. *Historical Geography*, *33*(1), 7-13. DeBats, D. A., & Gregory, I. N. (2011). Introduction to historical GIS and the study of urban history. *Social Science History*, *35*(4), 455-463. Rumsey, D., & Williams, M. (2002). *Historical maps in GIS* (pp. 1-17). na. Kucera, G. (2020). *Time in geographic information systems*. CRC Press. Wheatley, D. (2022). Cumulative viewshed analysis: A GIS-based method for investigating intervisibility, and its archaeological application. In *Archaeology and geographic information systems* (pp. 171-185). CRC Press. Kraak, M. J., & Ormeling, F. (2020). *Cartography: Visualization of geospatial data*. CRC Press.